FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – II
PAPER –2
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A *(01 – 04, 18 – 21, 35 – 38)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
  Section – A *(05 –07, 22 – 24, 39 – 41)*: This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
  Section – B *(08 – 13, 25 – 30, 42 – 47):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
  Section – C *(14 –17, 31 – 34, 48 – 51)*: This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

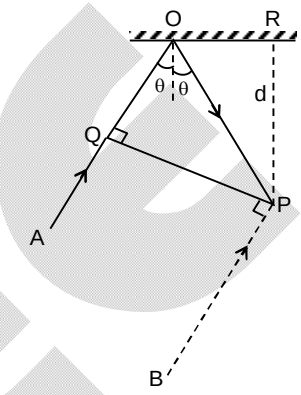
Physics

PART – I

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. In the given figure QP represents a plane wavefront and AO and BP, the corresponding extreme rays of monochromatic light of wavelength λ. The value of angle θ for which the ray BP and the reflected ray OP interfere constructively is given by

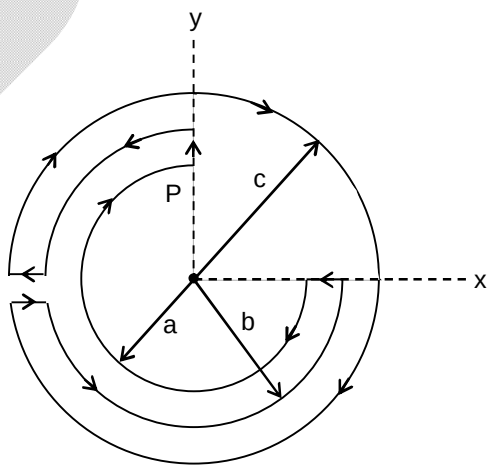


(A) $\cos\theta = \frac{\lambda}{2d}$ (B) $\cos\theta = \frac{\lambda}{4d}$

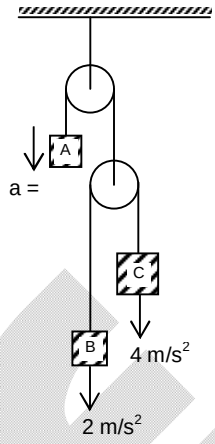
(C) $\sec\theta = \frac{\lambda}{3d}$ (D) $\sec\theta = \frac{2\lambda}{3d}$

2. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ, then θ is close to
(A) 1° (B) 15°
(C) 30° (D) 60°

3. For c = 2a and a < b < c, the magnetic field at point P will be zero when:
(A) a = b
(B) $a = \frac{3}{5}b$
(C) $a = \frac{5}{3}b$
(D) $a = \frac{1}{3}b$



4. In the figure shown acceleration of block B & C is 2 m/s^2 & 4 m/s^2 as shown in figure. Then acceleration of block A will be
(A) 3 m/s^2 upwards
(B) 3 m/s^2 downwards
(C) 6 m/s^2 upwards
(D) 6 m/s^2 downwards

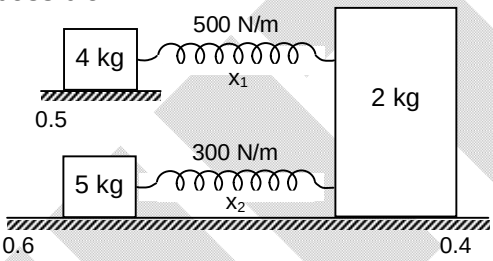


SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. The distance between two point objects P and Q is 32 cm. A convex lens of focal length 15 cm is placed between them so that the images of both the objects are formed at the same place. The distance of P from the lens could be
(A) 20 cm (B) 18 cm
(C) 16 cm (D) 12 cm

6. The system shown in is in equilibrium. Positive value of x_1 and x_2 denote extension of the springs while negative value denotes compressions. Which of the following is not possible?



(A) x_1 = 13 cm, x_2 = 2 cm (B) x_1 = 8 cm, x_2 = 3 cm
(C) x_1 = 1 cm, x_2 = –8 cm (D) x_1 = –1 cm, x_2 = –2 cm

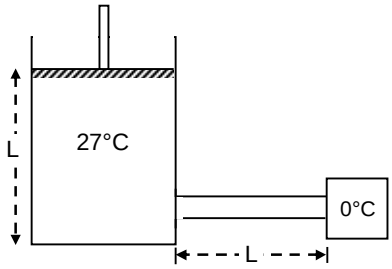
7. A traveling wave pulse is given by $y = \frac{6}{2+(x+3t)^2}$ where symbols have their usual meanings, *x, y* are in metre and t is in second. Then
(A) the pulse is traveling along +ve *x*-axis with velocity 3 m/s
(B) the pulse is traveling along –ve *x*-axis with velocity 3 m/s
(C) the amplitude of the wave pulse is 3 m
(D) the pulse is a symmetric about y - axis

SECTION – B

(Numerical Answer Type)

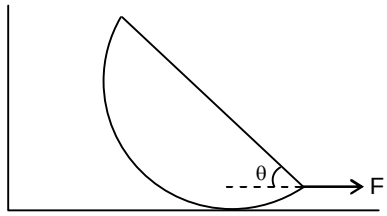
*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. 0.5 moles of an ideal gas at constant temperature 27°C kept inside a cylinder of length L and cross-section area A closed by a massless piston.



The cylinder is attached with a conducting rod of length L, cross section area = $\left(\frac{1}{9}\right)m^2$ and thermal conductivity K, whose other end is maintained at 0°C. If piston is moved such that rate of heat flow through the conducing rod is constant, then velocity of piston when It is at height L/2 from the bottom of cylinder is $\left(\frac{K}{PR}\right)m/sec$. Then find $\frac{P}{20}$.

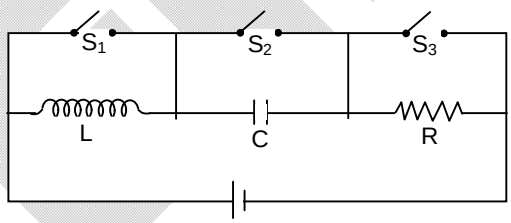
9. A solid hemisphere of radius r and mass M is pulled by means of a string (figure) so that it moves with a uniform velocity. If μ is the coefficient of friction with the surface, then the angle of inclination of the circular face of the hemisphere with the horizontal is given by



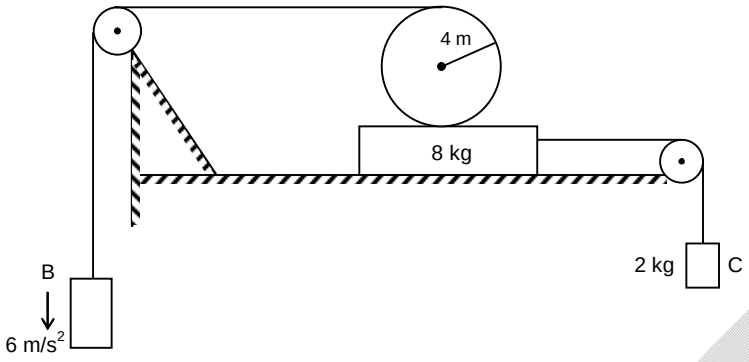
$\sin\theta = \frac{8\mu}{n+8\mu}$.

Find the value of 'n'?

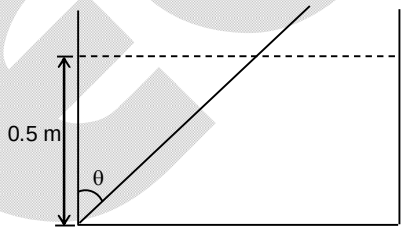
10. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take $\pi^2 = 10$)



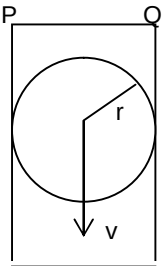
11. Figure shows a uniform smooth solid cylinder A of radius 4 m rolling without slipping on the 8 kg plank which, in turn, is supported by a fixed smooth surface. Block B is known to accelerate down with 6 m/s^2 and block C moves down with acceleration 2 m/s^2. What is the angular acceleration of cylinder. [g = 10 m/s^2]



12. A wooden plank of length 1 m and uniform cross section is hinged at one end to the bottom of a tank as shown in the figure. The tank is filled with water upto a height of 0.5 m. The specific gravity of the plank is 0.5. If the angle θ by the inclination of that plank makes with the vertical in the equilibrium position (exclude the case $\theta = 0$). Then the value of $\frac{1}{\cos^2\theta}$



13. A vertical ring of radius 'r' and resistance 'R' slips vertically between two frictionless and resistance less vertical rails. The rails which are joined at top there is uniform magnetic field B perpendicular to plane of ring and the rails. When speed of ring is v, current induced in section PQ is $\frac{nBrv}{R}$ find n



SECTION – C
(Numerical Answer Type)

This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 14 and 15

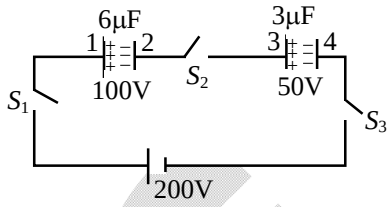
A light of wavelength λ is incident on a metal sheet of work function ϕ = 2eV. The wavelength λ varies with time as $\lambda = 3000 + 40t$, where λ is in Å and *t* is in second. The power incident on metal sheet is constant at 100 W. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of 3000Å. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1% (*hc* = 12400 eVÅ).

14. The time after which photomission will stop is (in sec.)

15. The total number of photoelectrons ejected in one hour is $X \times 10^{21}$. Find the value of 'X'.

Paragraph for Question Nos. 16 and 17

Two capacitors of capacity 6 μF and 3μF are charged to 100V and 50V separately and connected as shown in figure. Now all the three switches S_1, S_2 and S_3 are closed.



16. Charges on the 6μF capacitor in steady state will be (in μC)

17. Charges on the 3μF capacitor in steady state will be (μC)

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Chemistry

PART – II

SECTION – A
(One Options Correct Type)

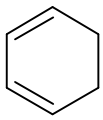
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

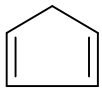
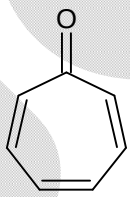
18. For a certain reaction of nth order

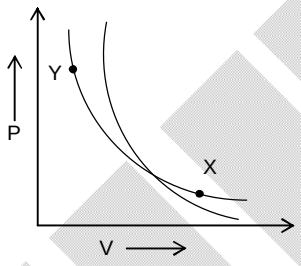
$t_{1/2} = \frac{2-\sqrt{2}}{k} C_0^{1/2}$, where k is rate constant & C_0 is initial concentration of reactant. The order of the reaction is

(A) 1 (B) 2
(C) 0 (D) 0.5

19. Which of the following molecule in pure form is unstable at room temperature?

(A)  (B) 

(C)  (D) 

20. 

The above P-V curves are for a gas under real and ideal behaviour.
(Except H_2 & He)
The value of compressibility factor varies on moving from point X to point Y as
(A) more than 1 to less than 1 (B) less than 1 to more than 1
(C) remains constant (D) can't predict

21. Hydrolysis of which of the following compounds is an example of disproportionation reaction.
(A) XeF_6 (B) XeF_4
(C) PCl_5 (D) PCl_3

SECTION – A

(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

22. Which of the following reaction(s) form(s) product(s) that show(s) stereoisomerism?

(A) $CH_3CH_2CHO \xrightarrow[NaCN]{HCN}$

(B) $CH_3CHO \xrightarrow[H^+(\text{catalyst})]{NH_2OH}$

(C) $CH_3CH_2COOH \xrightarrow{Cl_2/\text{red 'P'}}$

(D) $CH_3\underset{\underset{OH}{|}}{CH}COOH \xrightarrow{\text{Heating}}$

23. $FeCl_3 + K_4\left[Fe(CN)_6\right] \longrightarrow \text{Precipitate} \downarrow$

The correct statement(s) regarding the precipitate in above reaction is/are

(A) it is blue in colour

(B) oxidation number of Fe in product is +2 and +3

(C) the maximum van't Hoff factor of the precipitate can be seven

(D) the precipitate becomes soluble in hot water

24. Given $\Delta H_f^0(C_2H_6, g) = -85\,kJ/mole$,

$\Delta H_f^0(C_3H_8, g) = -104\,kJ/mole$,

$\Delta H_{sub}^0(C, s) = 718\,kJ/mole$ & B.E.of $(H-H) = 436$ kJ/mole.

Then, in kJ/mole, the

(A) C – C bond enthalpy is 218

(B) C – H bond enthalpy is 414

(C) C – C bond enthalpy is 345

(D) C – H bond enthalpy is 448

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. The following two reactions attain equilibrium in a container by taking initially only $FeSO_4$ and O_2 gas

$2FeSO_4(s) \rightleftharpoons Fe_2O_3(s) + SO_2(g) + SO_3(g); K_{P_1} = 900\,cm^2$ of Hg

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g); K_{P_2} = 25 \times 10^{-4}\,cm^{-1}$ of Hg

What is the equilibrium partial pressure of $O_2(g)$ in cm of Hg unit?

26. $KO_2(s) + CO_2(g) \longrightarrow P(s) + Q(g)$

$\xrightarrow{H_2O} R(soln) + S(soln) + Q(g)$

$\xrightarrow[\text{Starch}]{KI(soln)/H^+}$ Blue solution

What is the molar mass in g of the heaviest unknown product (out of P, Q, R and S) in above reactions?

27. $(P) \xrightarrow[\text{Alcohol, } \Delta]{KOH} (Q) \xrightarrow{Br_2/CCl_4} (R)$

(Alkyl chloride)

$CH_3 - \underset{\diagdown\ \diagup}{\underset{O}{CH}} - CH_2 \xleftarrow[-Br^-]{KOH(1\,eq)} (S) \xleftarrow[H_2O,\ S_N1]{KOH(1\,eq)}$ (R)

How many grams of (P) is required to form one mole of 2,3-dimethylbutane upon treatment with sodium metal in dry ether?

28. $HO - CH_2 - \underset{OH}{CH} - CH_2COOH \xrightarrow{SOCl_2(excess)} \text{Product}$

The molar mass in gram/mol of the organic product is y, then the value of 4y is

29. Cr(III) forms an octahedral complex with NH_3 and Cl ligands. The van't Hoff factor of the complex is 2 when it is 100% ionized in water. The complex ion shows geometrical isomerism. The molar mass of the complex in gram unit is x, then the value of 2x is
[Given mass of Cr = 52, N = 14, Cl = 35.5]

30. Calculate the reversible work done at 1000 K in cal unit by compressing one mole of an ideal gas to one-tenth $\left(\frac{1}{10}\right)$ of its original volume in an isothermal process?
[Use R = 2 cal K^{-1} mol^{-1}]

SECTION – C
(Numerical Answer Type)

This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 31 and 32

For dissociation reactions, like $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$, the degree of dissociation of SO_3 can be determined by vapour density measurement of the reacting species through the following relation.

$$\alpha = \frac{D - d}{(n-1)d}$$

D & d are Vapour density initially and at time 't' respectively

The molecular mass of the reaction mixture at equilibrium is 60. Initially only SO_3 was taken for the reaction. The vapour density 'D' is x.

Answer the following questions on the basis of the above write up.

31. The value of x/80 is

32. If vapour density of the mixture is 30, then value of 'α' is

Paragraph for Question Nos. 33 and 34

CH_3NH_2 is more basic than NH_3. The ionization constant (K_b) of CH_3NH_2 is $10^{-3.38}$. It reacts with HCl in the following way.

$$CH_3NH_2(aq) + HCl(aq) \rightleftharpoons CH_3NH_3^+(aq) + Cl^-(aq)$$

200 mL of 0.4 M solution of CH_3NH_2 is mixed with 200 mL of 0.2 M HCl solution. The mixture is stirred for complete reaction.
x = pH of the solution mixture and y = concentration of Cl^- ion in mol L^{-1} unit.

33. The value of x is

34. The value of y is

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Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. If sum $S=\sum_{r=1}^{2023}\frac{(-1)^{r+1}}{r}$, then S is also equal to

(A) $\sum_{r=1012}^{2023}\frac{(-1)^{r}}{r}$ (B) $\sum_{r=1012}^{2023}\left(\frac{1}{r}\right)$

(C) $\sum_{r=1012}^{2023}\frac{(-1)^{r+1}}{r}$ (D) $\sum_{r=1011}^{2023}\left(\frac{1}{r}\right)$

36. Number of integral solutions of the equation $\operatorname{sgn}\left(\sin^{-1}\left[\frac{\pi x}{6}\right]\right)=1$, where [x] denotes the greatest integer less than or equal to x and $\operatorname{sgn}(x)$ denotes signum function is

(A) 2 (B) 3
(C) 5 (D) 7

37. The line which contains all points (x, y, z) which are of the form $(x,y,z)=(2,-2,5)+\lambda(1,-3,2)$ intersects the plane $2x-3y+4z=163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a,b\in\mathbb{N}$ and $a>3$ then $(a+b)$ equals

(A) 23 (B) 95
(C) 27 (D) None of these

38. Consider the data with frequency distribution

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

The mean deviation about the median of the above data is

(A) 2.5 (B) 2.6
(C) 2.7 (D) 2.4

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. A line L passing through the point P (1, 4, 3), is perpendicular to both the lines $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$.

If the position vector of point Q on L is (a_1,a_2,a_3) such that $(PQ)^2=357$, then $(a_1+a_2+a_3)$ can be

(A) 16 (B) 15
(C) 2 (D) 1

40. Consider the system of equations $\sin x \cos 2y = \left(a^2 - 1\right)^2 + 1$ and $\cos x \sin 2y = a + 1$. Which of the following ordered pairs (x, y) of real numbers can satisfy the given system of equations for permissible real values of a?

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

(C) $\left(\frac{3\pi}{2}, \frac{-\pi}{2}\right)$ (D) $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$

41. Let $P(x)$ be a polynomial of n degree and $f(x) = \begin{cases} P\left(\frac{1}{x^3}\right) e^{-\frac{1}{x^4}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then

(A) $f(x)$ is discontinuous at $x = 0$ (B) $f(x)$ is continuous at $x = 0$

(C) $f(x)$ is non – differentiable at $x = 0$ (D) $f'(0) = \lim\limits_{x \to 0} f(x)$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. Let $f(x) = x + \sqrt{1 - x^2}$. Then the number of symmetric matrices A of order equals to the number of integers in the range of $f(x)$ and whose elements are the integral elements in the range of $f(x)$ and $\text{tr}(A)$ is equal to 1 or 0 is ____

43. $\lim\limits_{x \to 0} \frac{1}{\sin x} \int_0^{\ell n(1+x)} (1 - \tan 2y)^{\frac{1}{y}} dy = \frac{1}{e^n}$, then n is equal to ____

44. The area bounded by $y = 2 - |2 - x|$, and $y = \frac{3}{|x|}$ is $\frac{k - 3\ln 3}{2}$ then k is equal to ___

45. The curve which passes through the origin and the mid-point of the segment of its normal from foot of the normal to the x–axis lies on the parabola $2y^2 = x$ is $y^2 = \lambda x + 1 - e^{2x}$, then the value of λ is

46. Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x – axis. If $\tan\theta . \tan\phi = 2$, then the value of $2\alpha^2 - \beta^2$ is equal to

47. The slope of the line which belongs to the family of lines $(1 + \lambda)x + (\lambda - 1)y + 2(1 - \lambda) = 0 \left(\lambda \in \mathbb{R}\right)$ and makes shortest intercept on $x^2 = 4y - 4$ is

SECTION – C
(Numerical Answer Type)

This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 48 and 49

Two fixed points A and B are 4 units apart, and are on the same side of a moving line L. If perpendicular distances of A and B say p_1 and p_2 from the line L are such that $p_1 + 3p_2 = k$ (k being a constant), then the line L always touches a fixed circle C.

48. If the centre of the circle C is (α, β) then $\frac{\beta - \alpha}{8}$ is

49. If A and B are (–2, 0) and (2, 0) respectively, and the centre of the circle is on $5\lambda x^2 + y^2 - 7 = 0$ then λ is

Paragraph for Question Nos. 50 and 51

Let $f(x) = \left(1 + \frac{1}{x}\right)^x \ (x > 0)$ and $g(x) = \begin{cases} x\,\ell n\left(1 + \frac{1}{x}\right) & \text{if } 0 < x \\ 0 & \text{if } x = 0 \end{cases}$

50. $\lim\limits_{x \to \infty} \frac{2f(x)}{g(x)} =$

51. $\lim\limits_{n \to \infty} \left\{ f\left(\frac{1}{n}\right).f\left(\frac{2}{n}\right).f\left(\frac{3}{n}\right).....f\left(\frac{n}{n}\right) \right\}^{\frac{2}{n}}$ equals